

Qatar-1b

R-2739

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260618 · created 2026-09-02T01:13:25+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461209.8637 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.021
Transit depth (Rp/Rs)^2	1.59 +/- 0.53 [%]
Orbital Inclination (inc)	88.91 +/- 2.07
Airmass coefficient 1 (a1)	0.996 +/- 0.015
Airmass coefficient 2 (a2)	0.001 +/- 0.065
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	None
Optimal Aperture	3.57
Optimal Annulus	15.23
Transit Duration (day)	0.0798 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.126 ± 0.021 vs 0.1463 (deviation 0.97σ)
Duration fit vs published	0.0798 d vs 0.0692 d (deviation 3.04σ)
Mid-time O–C	-7.3 min
Depth SNR	6.0
Residual scatter	1.47 %

Light curve

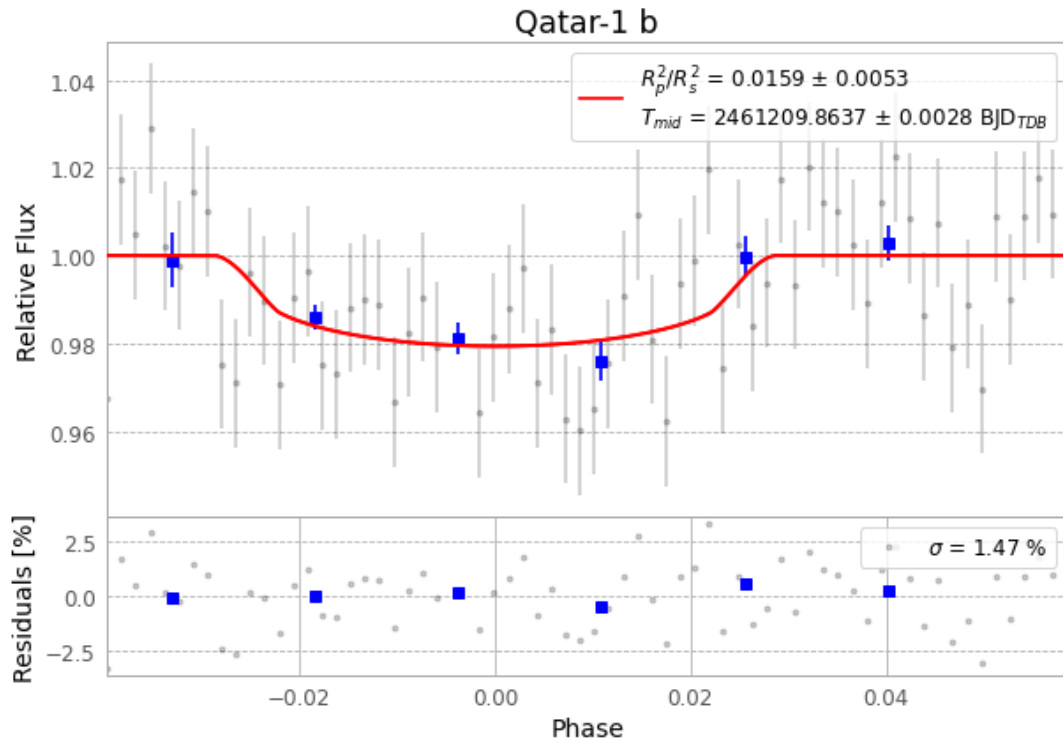


Figure 1 — FinalLightCurve_Qatar-1 b_2026-06-18.png (SHA-256 5b958029eff4d062...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1fc293d57f19339e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), Qatar-1260618072115.FITS → Qatar-1260618104215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260618.log (a6a7267b45a8...), FinalLightCurve_Qatar-1 b_2026-06-18.png (5b958029eff4...), FinalLightCurve_Qatar-1 b_2026-06-18.csv (d7ad47f514fe...)

Compute resources

Wall time 161.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:13Z.